Rendeiro 2020 CLL — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Judged on the regenerated embedding across a θ-sweep (1/2/3/5/10 × n_pcs 20/30/40) with per-neighbourhood iLISI. The malignant compartment is a per-patient archipelago: unintegrated kNN patient accuracy 99.7%, Leiden 99.3% patient-pure. Harmony mixes only by forced over-integration — even at θ=10, patient kNN accuracy stays 52% (chance 34%) and iLISI plateaus ~3.0/4. Signatures carry no manifold geography (Moran's I +0.02 = random-signature control). No single connected continuous developmental gradient — NO-GO.

Gate 2 — module geography → not evaluated

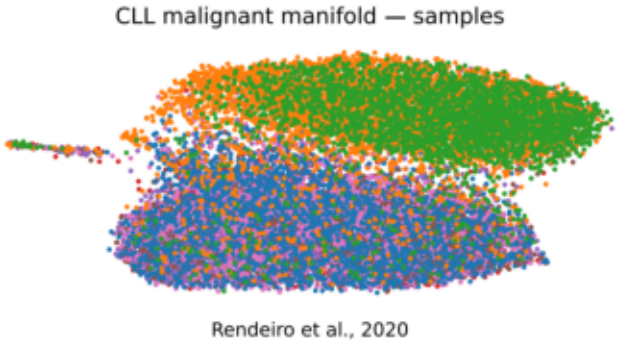
Not evaluated — the gate chain stops at the Gate 1 NO-GO (no integrated continuous manifold on which to read a Gast/pSom hub or arms). For the record, modules are near-zero and cell-cycle-orthogonal, and Gast–pSom co-vary (r=+0.05, no cross-repression).

Gate 3 — cross-repression → not evaluated

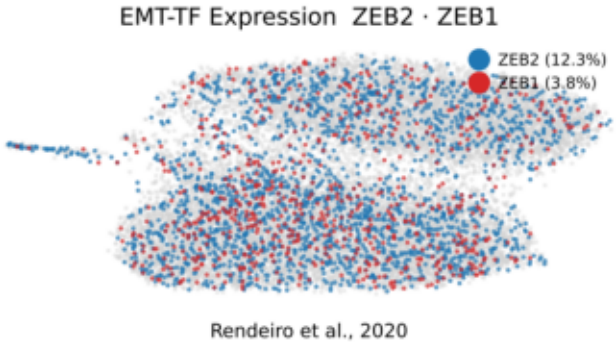
Not evaluated — reached only when a branching manifold clears Gates 1–2. No trajectory is fit (correlation / modules notebooks correctly not produced).

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
Chronic lymphocytic leukemia (Rendeiro et al., 2020), 10x — 18,598 malignant B-cells over 7 samples / 4 patients (longitudinal ibrutinib). Each patient is a genetically distinct CLL clone.

Verdict.
Pattern does NOT reproduce. Gate 1 NO-GO — a per-patient clonal archipelago with no shared continuous manifold and no embryonic module geography. No trajectory is warranted; analysis stops at preprocessing + the unconditional EMT-TF step.

Parameter choices

Min-cell threshold	≥100 (multiplicative cliff 1012→80, 12.65×); 7/11 samples	**Malignant cells**	18,854 raw → 18,657 (threshold) → 18,598 (island removal)
Overdispersed genes	528; PCA 50 comps, NO scaling	**Integration**	Harmony on sample θ=2 canonical; swept 1/2/3/5/10 × n_pcs 20/30/40
Embedding	n_pcs=20, n_neighbors=30, spread=1.3; 59 island cells removed	**Cell cycle**	42.2% cycling; modules cell-cycle-orthogonal (r +0.02) — no lobe to scope
Geometry	None — no manifold/trajectory fit (Gate 1 NO-GO); Gast–pSom +0.05 (not branching)	**EMT-TFs**	ZEB2 (12.3%) · ZEB1 (3.8%) — only TFs above 3% floor; VIM excluded